Lecture 2 student guide

How do forces and deformation become stress and strain?



The storyline: an apparatus applies and measures forces and displacements. Geometry converts them into stress and strain. Principal stresses describe the loading state, and Mohr analysis resolves that state onto any potential fracture plane.

What you should be able to do

- Distinguish quantities measured directly from quantities calculated using specimen geometry.
- Calculate nominal axial stress, axial strain, radial strain and small-strain volumetric strain.
- Recognize uniaxial, hydrostatic and conventional triaxial loading in principal-stress and p-q space.
- Resolve normal and shear stress on a plane and construct the corresponding Mohr circle.
- Explain how a strength envelope turns a stress calculation into a failure prediction.

Where to study in the book

READING	FOCUS	WHY IT MATTERS
Chapter 2, §2.1.1 pp. 5-7	Triaxial-test principle and terminology	Confining pressure, axial loading and differential stress
Chapter 2, §2.1.4 pp. 13-15	Measurements, control and data reduction	Force and displacement corrections; nominal stress and axial strain
Chapter 3, §§3.2-3.3 pp. 18-27	Fracture types and observed failure stress	Pressure dependence, Mohr envelope and Coulomb criterion
Optional: Chapter 2, §§2.1.2-2.1.3 pp. 7-13	Apparatus and specimen jacketing	Useful experimental context; not required for the calculations

Reference: M. S. Paterson & T.-f. Wong, Experimental Rock Deformation - The Brittle Field, 2nd ed. (2005). Page numbers refer to the printed book.

1. From measurements to material response

Always write the measured quantity first, then the geometry used to normalize it.

MEASURED DIRECTLY	CALCULATED
Force F [N] Axial/radial displacement [mm] Initial diameter and length [mm] Confining pressure P_c [MPa]	Stress [MPa] Axial and radial strain [-] Area A_0 [mm²] Principal, mean and differential stresses [MPa]

NOMINAL AXIAL STRESS

$$\sigma_a = F / A_0 \qquad A_0 = \pi D_0^2 / 4$$

Use the original area when reporting nominal stress. 1 N/mm² = 1 MPa.

AXIAL STRAIN

$$\varepsilon_a = (L_0 - L) / L_0 = \Delta L / L_0$$

Compression-positive convention: shortening gives $\varepsilon_a > 0$.

RADIAL STRAIN

$$\varepsilon_r = (D_0 - D) / D_0$$

With this convention, radial widening gives $\varepsilon_r < 0$.

VOLUMETRIC STRAIN

$$\varepsilon_v = \varepsilon_a + 2\varepsilon_r$$

Small strain and axisymmetric deformation. Positive ε_v means volume decrease.

Three comparisons worth remembering

SAME INPUT	WHAT CHANGES?	REASON
Equal force	Smaller area → larger stress	$\sigma = F/A$
Equal shortening	Shorter sample → larger strain	$\varepsilon = \Delta L/L_0$
Equal stress and equal E	Equal strain, but longer sample shortens more	$\varepsilon = \sigma/E$; $\Delta L = \varepsilon L_0$

Experimental caution: $\delta_{machine} = \delta_{specimen} + \delta_{apparatus}$. Frame compliance and seating can make machine travel larger than true specimen shortening.

2. Loading paths and stress on a plane

The boundary condition determines the stress path; plane orientation determines how that stress state is resolved.

TEST	PRINCIPAL STRESSES	IN p-q SPACE
Uniaxial compression	$\sigma_1 > 0; \sigma_2 = \sigma_3 = 0$	$q = 3p$
Hydrostatic loading	$\sigma_1 = \sigma_2 = \sigma_3 = P$	q = 0; p increases
Conventional triaxial compression	$\sigma_1 > \sigma_2 = \sigma_3 = P_c$	at constant P_c: $p = P_c + q/3$

Stress invariants used here: $p = (\sigma_1 + 2\sigma_3)/3$ and $q = \sigma_1 - \sigma_3$. Repeated triaxial tests at different P_c locate multiple failure points and build a failure envelope.

Angle convention - draw the normal first

θ = angle between plane normal **n** and σ_1
β = angle between the physical plane and σ_1
θ + β = 90°

Do not interchange θ and β.
Every transformation equation below uses θ, the angle of the normal.

STRESS TRANSFORMATION

$$\sigma_n = \sigma_1 \cos^2\theta + \sigma_3 \sin^2\theta$$
$$|\tau| = (\sigma_1 - \sigma_3) \sin\theta \cos\theta$$

MOHR-CIRCLE FORM

$$C = (\sigma_1 + \sigma_3)/2$$
$$R = (\sigma_1 - \sigma_3)/2$$
$$\sigma_n = C + R \cos(2\theta)$$
$$|\tau| = R \sin(2\theta)$$

SPECIAL ORIENTATIONS

$\theta = 0° \rightarrow (\sigma_n, |\tau|) = (\sigma_1, 0)$

$\theta = 45° \rightarrow (\sigma_n, |\tau|) = (C, R)$

$\theta = 90° \rightarrow (\sigma_n, |\tau|) = (\sigma_3, 0)$

Geometric memory: the physical plane normal rotates by θ while the point on the Mohr circle moves through 2θ.

3. From stress state to failure

Mohr analysis calculates the stresses on a plane. A strength criterion is still needed to decide whether that plane fails.

MOHR-COULOMB FIRST APPROXIMATION

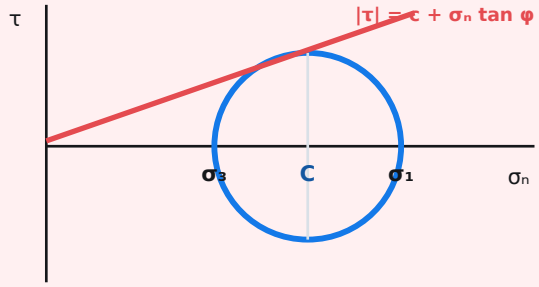
$$|\tau| = c + \sigma_n \tan \varphi$$

c is cohesion; **φ** is the friction angle. Failure begins when the Mohr circle first touches the strength envelope.

For triaxial compression, with this lecture's angle convention:

$\beta_f = 45° - \varphi/2$ and $\theta_f = 45° + \varphi/2$

Real rock envelopes may curve with pressure; the straight line is a useful local approximation.



Test yourself before reading the hints

1. Two samples carry 40 kN. Their areas are 500 and 1000 mm². What are their nominal axial stresses?

2. Two samples shorten by 0.10 mm. Their initial lengths are 50 and 100 mm. What are their axial strains?

3. If $\varepsilon_a = 0.002$ and $\varepsilon_r = -0.0005$, calculate ε_v. Does volume increase or decrease?

4. Why can a specimen widen even though the lateral surfaces are stress-free in a uniaxial test?

5. For $\sigma_1 = 100$ MPa and $\sigma_3 = 20$ MPa, find the Mohr-circle centre, radius and maximum shear stress.

6. Tests A (80, 0 MPa) and B (100, 20 MPa) have the same differential stress. Why are their stress states not identical?

7. Explain why machine travel may overestimate specimen shortening. Where should displacement be measured?

8. In one sentence: why does vertical loading commonly produce an inclined shear fracture?

Answer hints: 1) 80 and 40 MPa. 2) 0.002 (0.20%) and 0.001 (0.10%). 3) $\varepsilon_v = 0.001$: volume decreases. 4) Zero lateral traction does not mean zero lateral strain. 5) C = 60 MPa, R = $|\tau|_{max}$ = 40 MPa. 6) Their circle centres and mean stresses differ: p = 26.7 and 46.7 MPa. 7) Frame compliance and seating contribute; measure as close to the specimen as possible. 8) An inclined plane can reach the combined normal/shear condition required by the strength envelope.

A SIMPLE 30-MINUTE REVIEW	BEFORE YOU MOVE ON
10 min: redraw the measurement → stress/strain chain. **10 min:** construct one Mohr circle without notes. **10 min:** answer questions 4, 6 and 8 aloud.	☐ I can state the sign convention. ☐ I can distinguish force from stress and shortening from strain. ☐ I draw the plane normal before choosing θ. ☐ I know that calculation alone does not predict failure.